SAKIB HOSSAIN

COMPUTER SCIENCE & TECHNOLOGY STUDENT | SOFTWARE DEVELOPMENT | AI | WEB | EMBEDDED SYSTEMS

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PROFESSIONAL SUMMARY

Computer Science and Technology student with a multidisciplinary background spanning software development, AI-enabled applications, web development, and embedded systems. Builds software and hardware prototypes using Python, C/C++, JavaScript, Flask, Tkinter, SQLite, microcontrollers, and sensor interfaces. Project work includes Gemini-integrated desktop and web applications, an ESP32-based sensor-monitoring prototype, and Arduino-based robotics exploration. Combines technical study with a completed Bachelor of Business Studies degree, self-directed learning, and participation in national and institutional technical events.

TECHNICAL SKILLS

Programming and Software Development: Python, C/C++, JavaScript, HTML5, CSS3, Git, GitHub

Web and Application Development: Flask, Tkinter, SQLite, frontend development, desktop application development, API integration

Artificial Intelligence: Gemini API, AI-powered application development, AI integration, AI-assisted learning applications

Embedded Systems, IoT and Robotics: ESP32, Arduino, BBC micro:bit, microcontroller programming, sensor interfacing, hardware-software integration, IoT development, robotics, electronic prototyping, motor control, hardware debugging

Hardware and Control: IR/optical sensor arrays, pulse oximeter sensors, galvanic sensors, DC motors, servo motors, H-bridge motor drivers (including L298N), PID-based motor control

SELECTED PROJECTS

Advanced Health Monitoring System
Technologies: ESP32, pulse oximeter sensor, galvanic sensor, HTML, CSS, JavaScript, Flask, ngrok
Developed an IoT-based health-monitoring prototype that combines an ESP32 with pulse-oximeter and galvanic sensor inputs. Presented sensor information through a web interface built with HTML, CSS, JavaScript, and Flask; used ngrok as part of the development workflow.

Personal AI Tutor
Technologies: Python, Tkinter, Gemini API
Built a desktop-based AI tutor that helps students understand computer science and programming concepts through guided explanations, debugging support, and problem-solving prompts. Designed around a teaching-first approach, it explains concepts and helps learners reason through issues rather than supplying ready-made code.

AIDuck
Technologies: HTML, CSS, JavaScript, Flask, SQLite, Gemini API
Developed an AI-powered web application using a browser-based frontend, a Flask backend, SQLite data storage, and Gemini API integration. The project brings together frontend development, server-side application logic, database use, and AI service integration.

EDUCATION

Diploma in Engineering - Computer Science & Technology
A K Khan UCEP Polytechnic Institute, Chattogram | Currently enrolled - 4th Semester

Bachelor of Business Studies (BBS) - Degree Pass Course
Hazera-Taju Degree College, Chattogram | National University, Bangladesh | Completed

COMPETITION AND TECHNICAL EVENT PARTICIPATION

- National Robotics Championship (NRC 2025) - Line Following Robot.
- Chittagong Science Carnival 4.0 - Project Showcase: IoT-Based Smart Traffic Management System.
- AKKUPI x RoboStrom - Mini Soccer.
- DUET TechFest 2025 - Line Following Robot.
- EEE Day 3.0 (2026) - Line Following Robot.

TECHNICAL LEARNING AND KNOWLEDGE SHARING

- Embedded systems and robotics practice: Public GitHub documentation describes an Arduino-compatible line-following robot using a five-channel IR/optical sensor array, PID-based motor control, two DC motors, an H-bridge motor driver, and an auxiliary servo mechanism.
- Technical knowledge sharing: Published a beginner-oriented explanation of proportional, integral, and derivative control, relating PID tuning to line-following accuracy, stability, and smoother robot movement.
- Python and data-science learning: Shared learning updates on Python for Data Science and introductory Pandas practice, including DataFrame creation from lists, dictionaries, and CSV files.
- AI education and application development: Built and publicly documented a Gemini-powered programming tutor with a focus on conceptual understanding, debugging guidance, and responsible learning rather than code copy-pasting.
- Professional development: Participated in the Computer Science Career Guideline Event 2026 at A K Khan UCEP Polytechnic Institute on 15 October 2026, engaging with presentations on software engineering, system design, web and mobile development, and technical career pathways.
- Ongoing technical exploration: Continues hands-on learning across software development, AI, robotics, embedded systems, sensor interfacing, and hardware-software integration; shares selected learning progress through public professional posts and project repositories.

PROFESSIONAL INTERESTS

Software engineering, AI-enabled applications, web development, desktop application development, embedded software, robotics, IoT systems, and practical hardware-software integration.